

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 03 · Kinematics Graphs

Questions

Student book · questions with writing space

- 55 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 03

Kinematics Graphs

Your classified route

14

QUESTIONS

150

MARKS

2

FAMILIES

6

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 03 | FAMILY 01

Interpret and transform motion graphs

WHAT THIS FAMILY ASKS

Use gradients, signed/unsigned areas and representation changes among s-t, v-t, speed-t and a-t graphs.

VARIANT ROUTE

V01 Read gradients and areas

V02 Convert between graph types

V03 Find an unknown from graph area

9 classified questions

90 total marks

6.

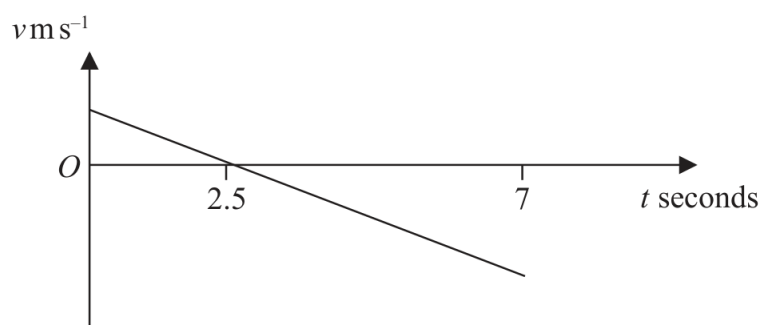


Figure 3

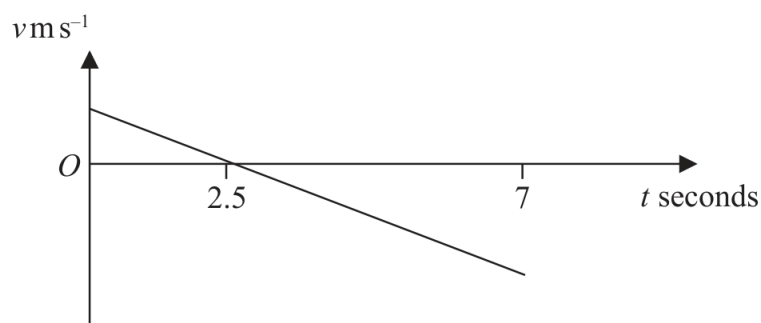
A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown.

(2)

**Figure 3**

A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(b) Find the height of A above the ground.

(3)

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2.

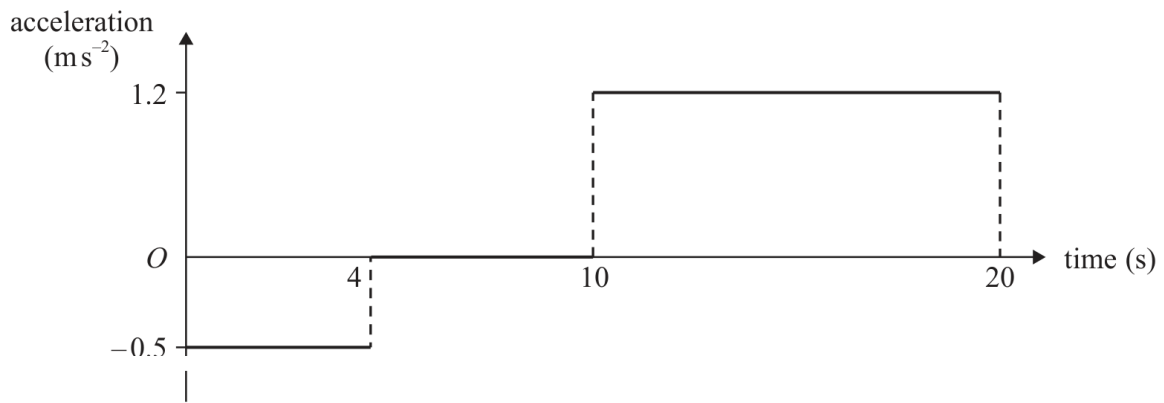


Figure 2

Two fixed points, A and B , are on a straight horizontal road.

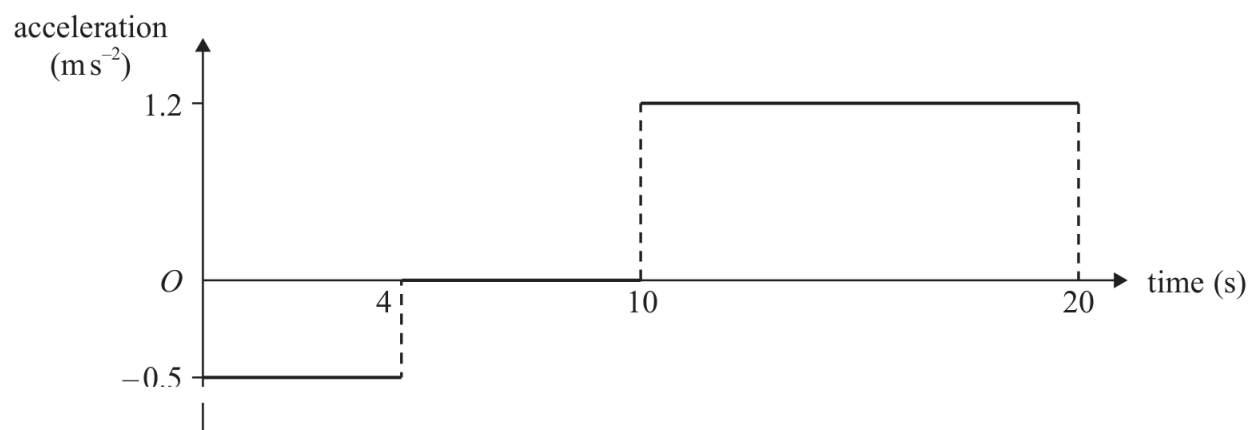
The **acceleration-time** graph in Figure 2 represents the motion of a car travelling along the road as it moves from A to B .

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ m s}^{-1}$

(a) Show that $v = 18$

(3)

**Figure 2**

At time $t = 0$, the car passes through A with speed 8 ms^{-1}

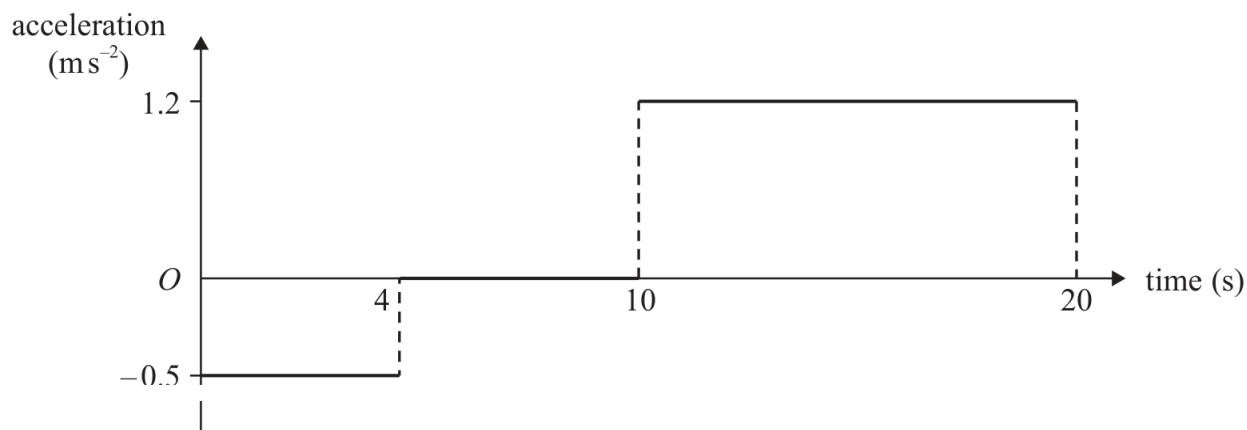
At time $t = 20 \text{ s}$, the car passes through B with speed $v \text{ ms}^{-1}$

$$v = 18$$

(b) Sketch a speed-time graph for the motion of the car from A to B .

(3)

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**Figure 2**

At time $t = 0$, the car passes through A with speed 8 m s^{-1}

$v = 18$

(c) Find the distance AB .

(4)

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6.

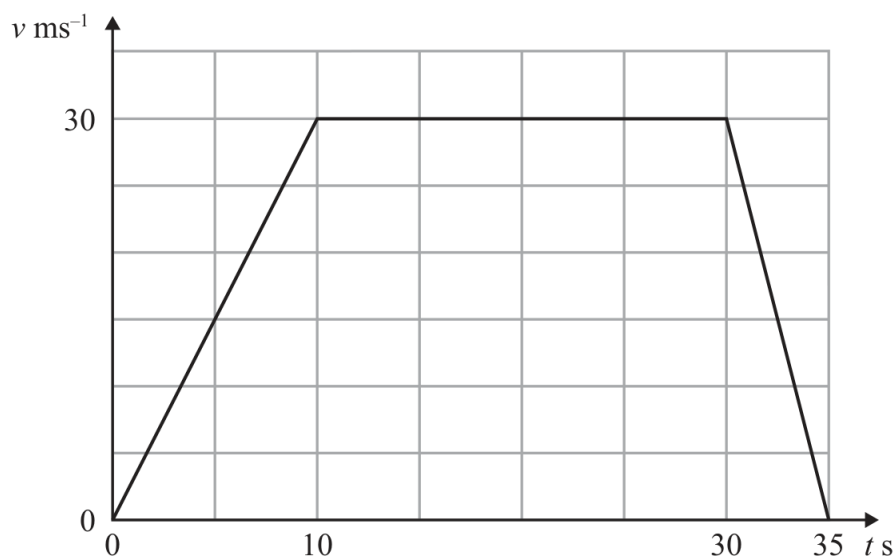


Figure 4

(a) Find

- (i) the acceleration of the car for the period $0 \leq t \leq 10$
- (ii) the deceleration of the car for the period $30 \leq t \leq 35$

(2)

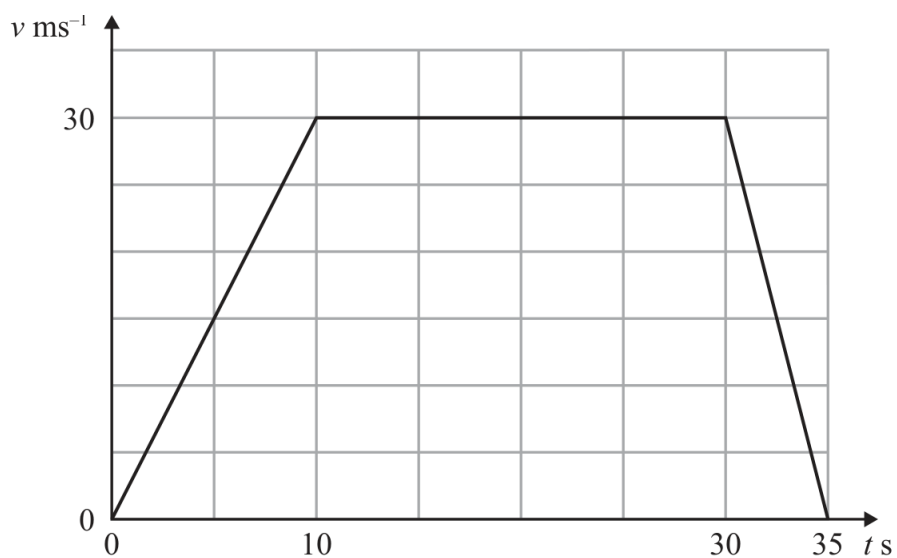
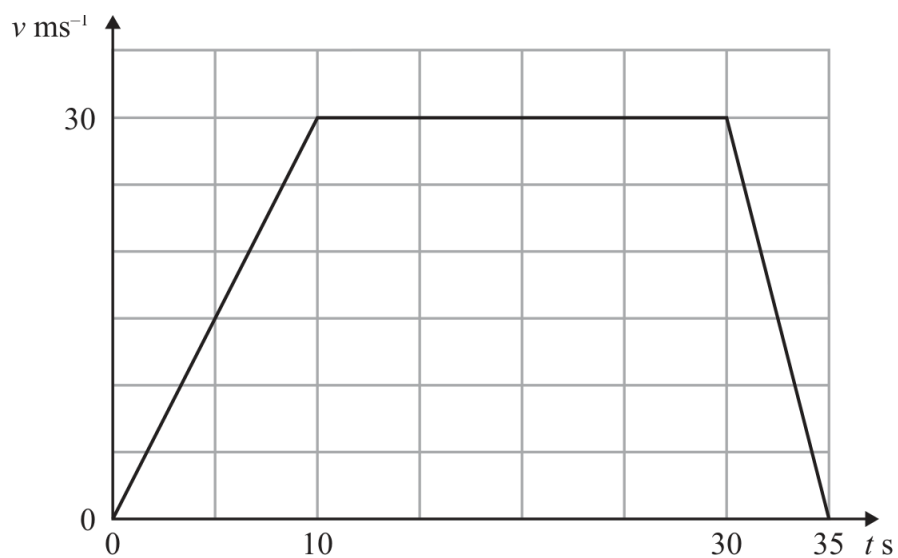


Figure 4

(b) Sketch an acceleration-time graph for the car for the period $0 \leq t \leq 35$

(2)

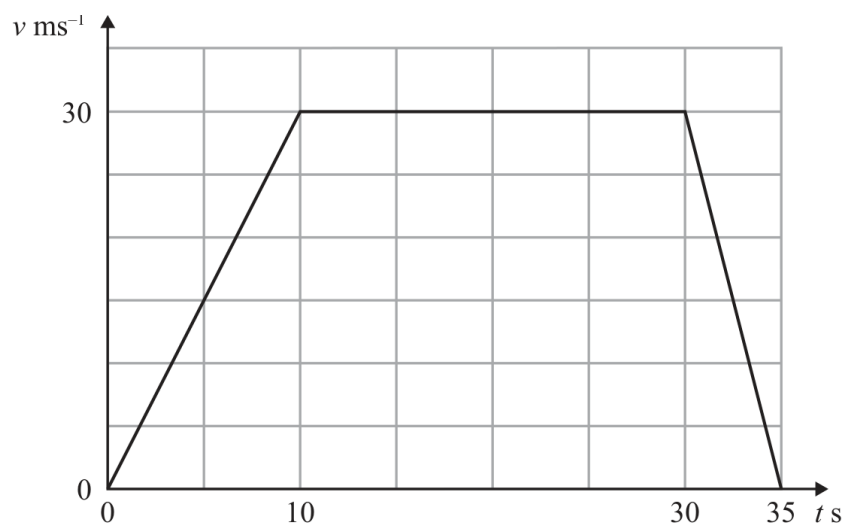
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**Figure 4**

(c) Find the distance travelled by the car for the period $0 \leq t \leq 35$

(2)

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When $t = 5$, a motorcycle starts from rest at O .

The motorcycle travels along the same road as the car and in the same direction.

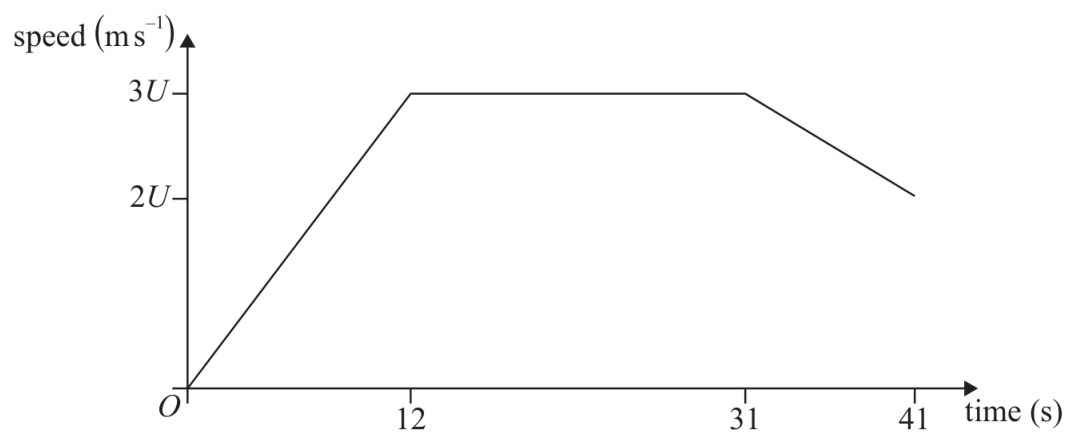
For the period $5 \leq t \leq 20$, the acceleration of the motorcycle is $A \text{ ms}^{-2}$, where A is a positive constant.

The motorcycle catches up with the car when $t = 20$

(d) Find the value of A .

(4)

2.

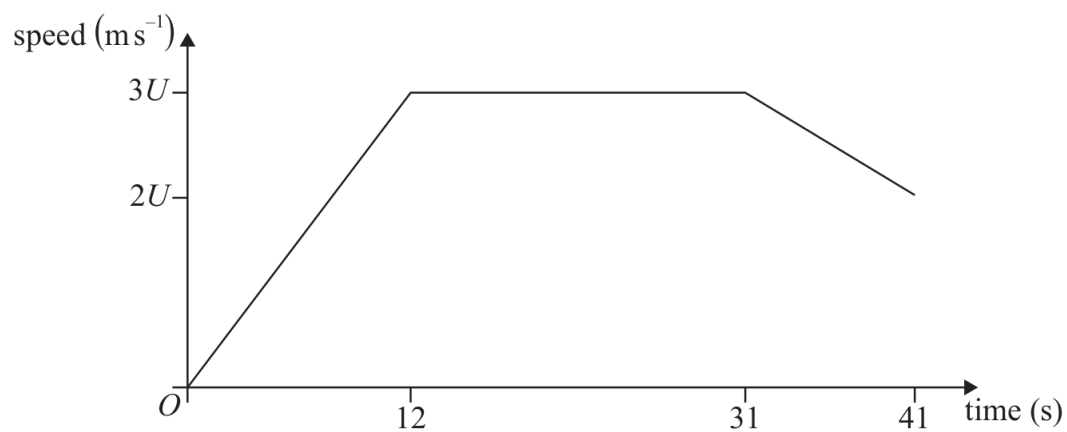


The distance travelled by P between $t = 0$ and $t = 41$ s is 600 m.

(a) Show that $U = 6$

(3)

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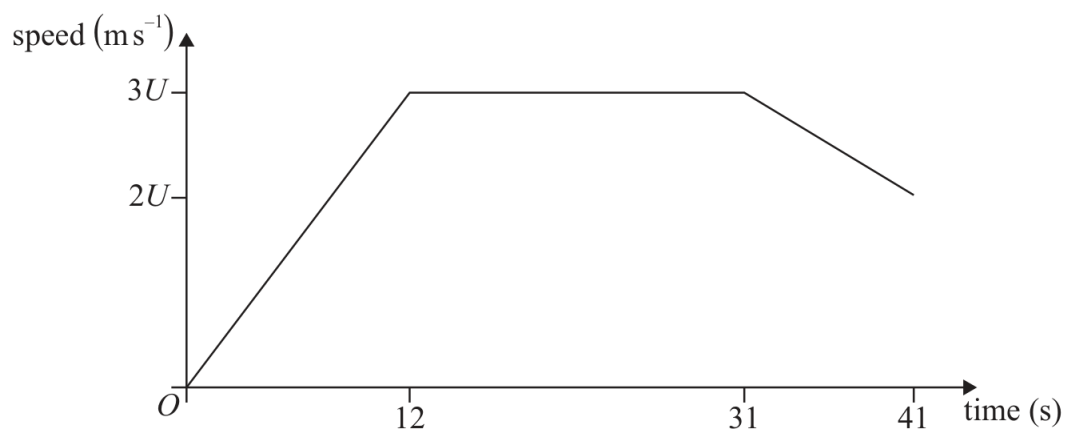


$$U = 6$$

- (b) Find the distance travelled by P whilst moving from rest to a speed of 6 ms^{-1}

(2)

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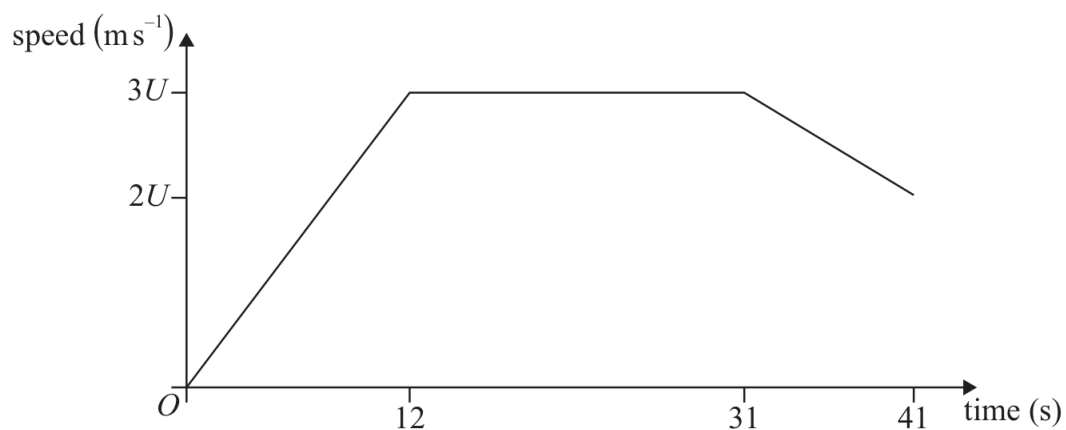


$$U = 6$$

- (c) Find the acceleration of P between $t = 0$ and $t = 12$ s.

(1)

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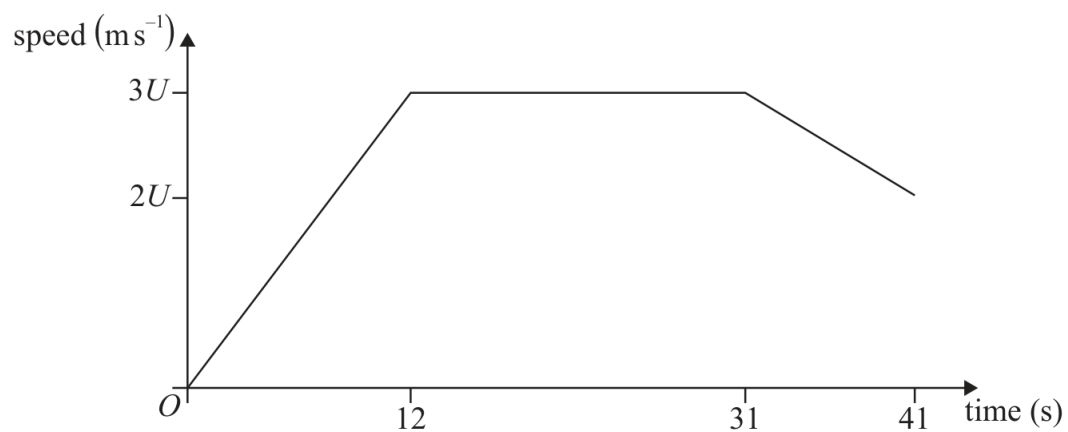


$$U = 6$$

- (d) Find the deceleration of P between $t = 31$ s and $t = 41$ s.

(2)

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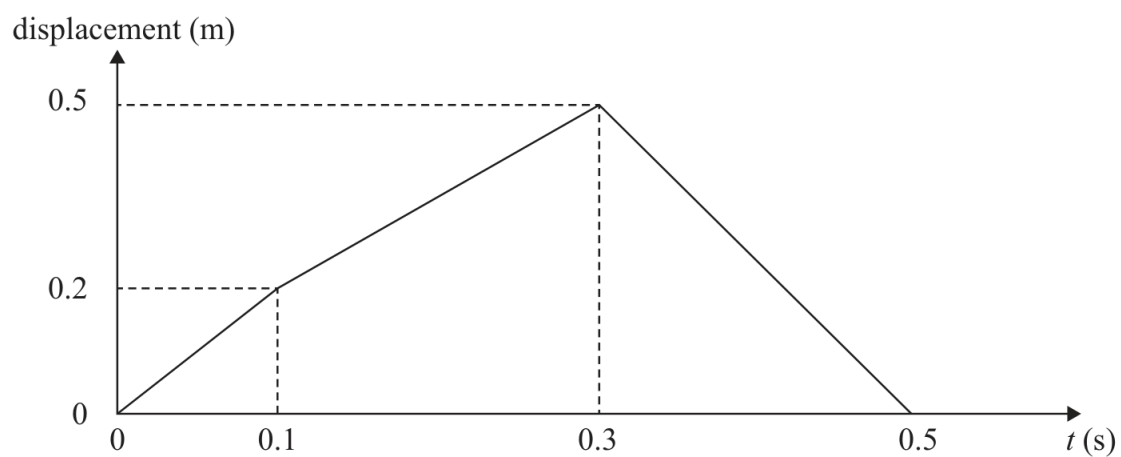


(e) Sketch an acceleration-time graph to represent the motion of P from $t = 0$ to $t = 41$ s.

(2)

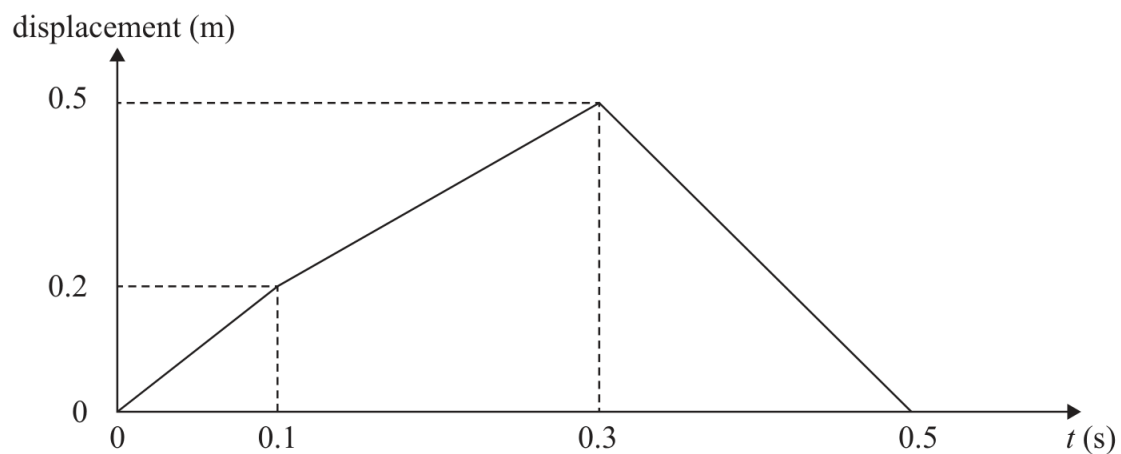
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3.

**Figure 2**

(a) State the displacement of the ball from O at $t = 0.5$

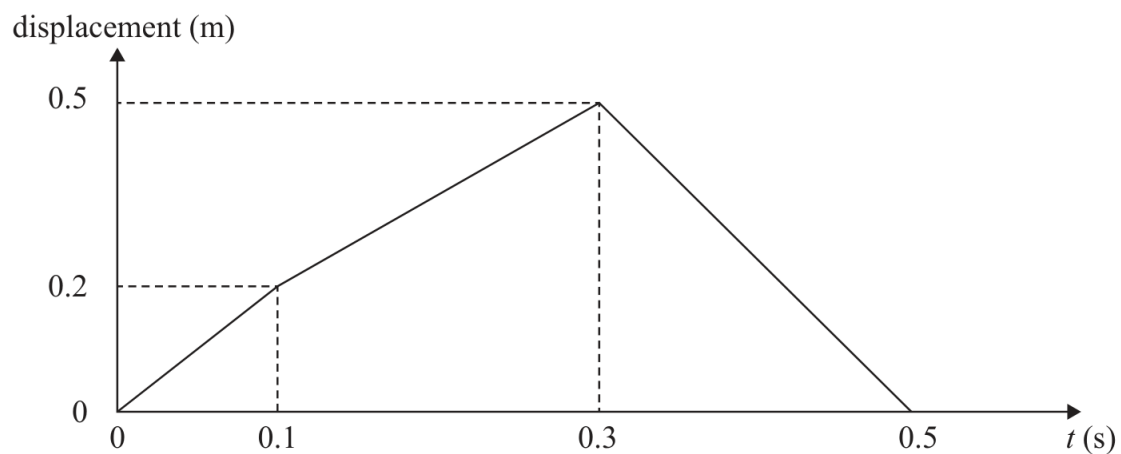
(1)

**Figure 2**

(b) Find the total distance moved by the ball between $t = 0$ and $t = 0.5$

(1)

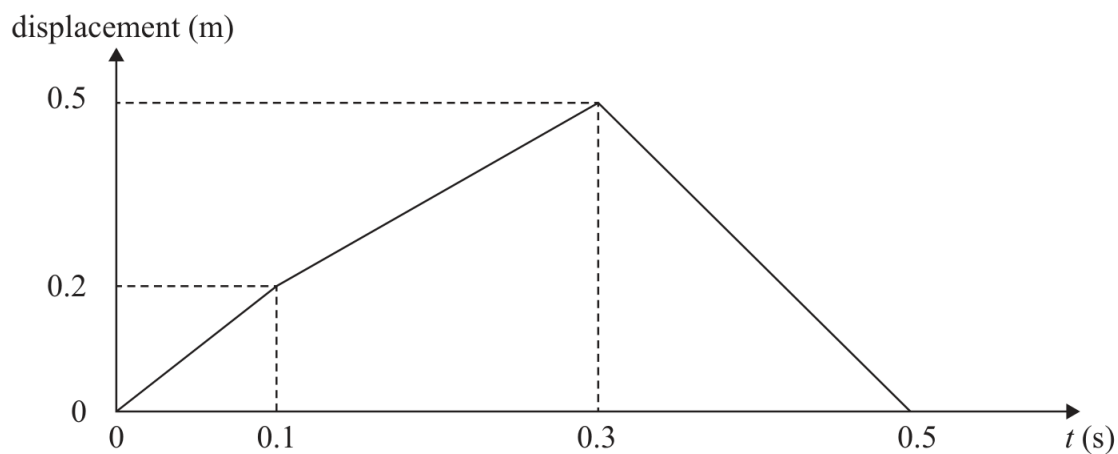
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**Figure 2**

(c) Sketch a velocity-time graph for the motion of the ball from $t = 0$ to $t = 0.5$

(4)

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**Figure 2**

A small ball of mass 0.04 kg moves along a straight line on a pinball table.

- (d) Find the magnitude of the impulse exerted on the ball at $t = 0.3$, stating the units of your answer.

(4)

6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (a) Sketch a speed-time graph for the model of the motion of the athlete from the instant when she starts to the instant when she crosses the finishing line.

(2)

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6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (b) Write down an expression for V in terms of T .

(1)

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6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (c) Show that $T^2 - kT + 500 = 0$, where k is a constant to be found.

(4)

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The total time from when the athlete starts to when she crosses the finishing line is 30 s.

$$T^2 - kT + 500 = 0,$$

(d) Hence find the value of T , justifying your answer carefully.

(3)

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6. An athlete runs a 200 m race along a straight horizontal track.

In a model of the motion of the athlete, air resistance is ignored, the athlete starts from rest at time $t = 0$ seconds and moves with uniform acceleration 0.8 m s^{-2} for T seconds, reaching a speed of $V \text{ m s}^{-1}$. She then maintains this speed until she crosses the finishing line.

The total time from when the athlete starts to when she crosses the finishing line is 30 s.

- (e) Considering your speed-time graph or otherwise, state two ways, apart from including air resistance, in which the model could be made to be more realistic.

(2)

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7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens.

(1)

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The parachutist is modelled as a particle.

- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he hits the ground.

(2)

DR ESLAM AHMED

7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (c) Find the value of T .

(5)

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The parachutist is modelled as a particle.

- (d) Find, to the nearest metre, the height of the point H above the ground.

(4)

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1. A train travels along a straight horizontal track between two stations A and B .

The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station B .

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

- (a) On the axes below, sketch a speed–time graph to illustrate the motion of the train as it moves from station A to station B .



1. A train travels along a straight horizontal track between two stations A and B .

The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

The train then travels at a constant speed of 20 m s^{-1} for 3 minutes before decelerating uniformly until it comes to rest at station B .

The magnitude of the acceleration of the train is twice the magnitude of the deceleration.

Stations A and B are 4.8 km apart.

- (b) Find the value of T

(5)

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The train starts from rest at station A and accelerates uniformly for T seconds until it reaches a speed of 20 m s^{-1}

(c) Find the acceleration of the train during the first T seconds of its motion.

(2)

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2. The fixed points A , B and C lie in a straight line on a horizontal road.

- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
- From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
- The motorbike takes T_1 seconds to travel from A to B
- From B , the motorbike decelerates uniformly until it comes to rest at C
- The motorbike takes T_2 seconds to travel from B to C

(a) Sketch a speed-time graph for the motion of the motorbike as it moves from A to C .

(3)

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2. The fixed points A , B and C lie in a straight line on a horizontal road.
- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
 - From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
 - The motorbike takes T_1 seconds to travel from A to B
 - From B , the motorbike decelerates uniformly until it comes to rest at C
 - The motorbike takes T_2 seconds to travel from B to C

The distance AB is 132 m and the distance BC is 136 m.

(b) Find, in terms of V , an expression for

(i) T_1

(ii) T_2

(4)

DR ESAM AHMED

2. The fixed points A , B and C lie in a straight line on a horizontal road.
- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
 - From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
 - The motorbike takes T_1 seconds to travel from A to B
 - From B , the motorbike decelerates uniformly until it comes to rest at C
 - The motorbike takes T_2 seconds to travel from B to C

The distance AB is 132 m and the distance BC is 136 m.

Given that the motorbike takes 28 s to travel from A to C ,

(c) find the value of V ,

(2)

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- The motorbike takes T_1 seconds to travel from A to B
- From B , the motorbike decelerates uniformly until it comes to rest at C
- The motorbike takes T_2 seconds to travel from B to C

The distance AB is 132 m and the distance BC is 136 m.

(d) find the deceleration of the motorbike.

(2)

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TOPIC 03 | FAMILY 02

Two-mover graph comparison and chase

WHAT THIS FAMILY ASKS

Compare distances/displacements of two movers represented on a shared or staggered time axis.

VARIANT ROUTE**V01** Staggered start**V02** Overtake or catch-up**V03** Two profiles constrained to finish together**5 classified questions**

60 total marks

8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (a) Sketch, on the same axes, a speed-time graph for the motion of tram A and a speed-time graph for the motion of tram B , from $t = 0$ to the instant when tram B overtakes tram A .

(3)

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8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

At the instant when the two trams are moving with the same speed, tram A is d metres in front of tram B .

- (b) Find the value of d .

(5)

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8. Two trams, tram A and tram B , run on parallel straight horizontal tracks. Initially the two trams are at rest in the depot and level with each other.

At time $t = 0$, tram A starts to move. Tram A moves with constant acceleration 2 m s^{-2} for 5 seconds and then continues to move along the track at constant speed.

At time $t = 20$ seconds, tram B starts from rest and moves in the same direction as tram A . Tram B moves with constant acceleration 3 m s^{-2} for 4 seconds and then continues to move along the track at constant speed.

The trams are modelled as particles.

- (c) Find the distance of the trams from the depot at the instant when tram B overtakes tram A .

(5)

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5.

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ m s}^{-1}$

The car then travels at a constant speed of $V \text{ m s}^{-1}$ until time $t = T$ seconds.

- (a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

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A motorbike also travels along the same road.

- The motorbike starts from rest at time $t = 10\text{ s}$ and accelerates uniformly for 40 s
 - The acceleration of the motorbike is the **same** as the acceleration of the car
 - The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V\text{ m s}^{-1}$ at time T seconds
- (b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

(2)

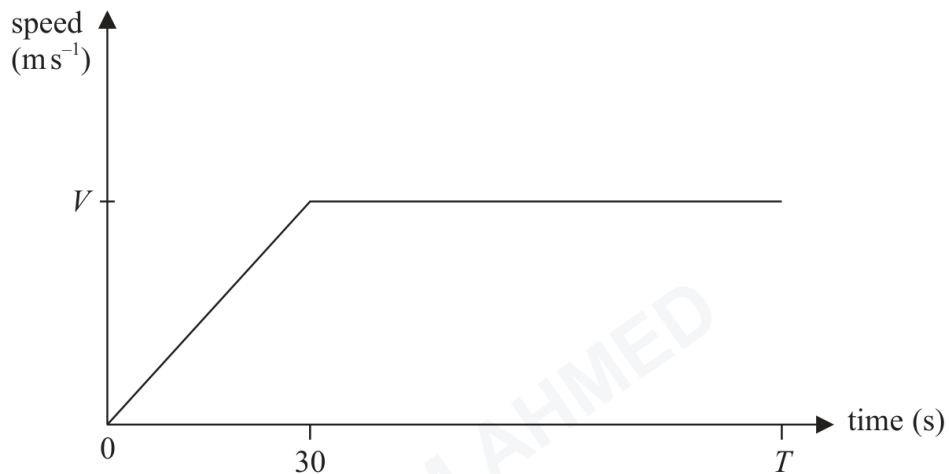


Figure 2

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ m s}^{-1}$

The car then travels at a constant speed of $V \text{ m s}^{-1}$ until time $t = T$ seconds.

- The motorbike starts from rest at time $t = 10 \text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V \text{ m s}^{-1}$ at time T seconds

(c) Show that the constant speed of the motorbike is $\frac{4V}{3} \text{ m s}^{-1}$

(2)

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The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V \text{ m s}^{-1}$

The car then travels at a constant speed of $V \text{ m s}^{-1}$ until time $t = T$ seconds.

- The motorbike starts from rest at time $t = 10 \text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V \text{ m s}^{-1}$ at time T seconds

$$\frac{4V}{3} \text{ m s}^{-1}$$

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

(d) Find the value of T

(5)

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4. The points A and B lie on the same straight horizontal road.

Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$

He then travels at constant speed $V \text{ m s}^{-1}$

When $t = 42$, cyclist P passes B .

Given that the distance AB is 120 m,

- (a) show that $V = 3.2$

(3)

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Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$

He then travels at constant speed $V \text{ m s}^{-1}$

When $t = 42$, cyclist P passes B .

Given that the distance AB is 120 m.

(b) Find the acceleration of cyclist P between $t = 0$ and $t = 9$

(2)

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Cyclist P continues to cycle along the road in the same direction at the same constant speed, $V \text{ m s}^{-1}$

When $t = 6$, a second cyclist Q sets off from A and travels in the same direction as P along the same road. She accelerates for T seconds until her speed is 3.6 m s^{-1}

She then travels at constant speed 3.6 m s^{-1}

Cyclist Q catches up with P when $t = 54$

- (c) On Figure 2, on page 11, sketch a speed-time graph showing the journeys of **both** cyclists, for the interval $0 \leq t \leq 54$

(3)

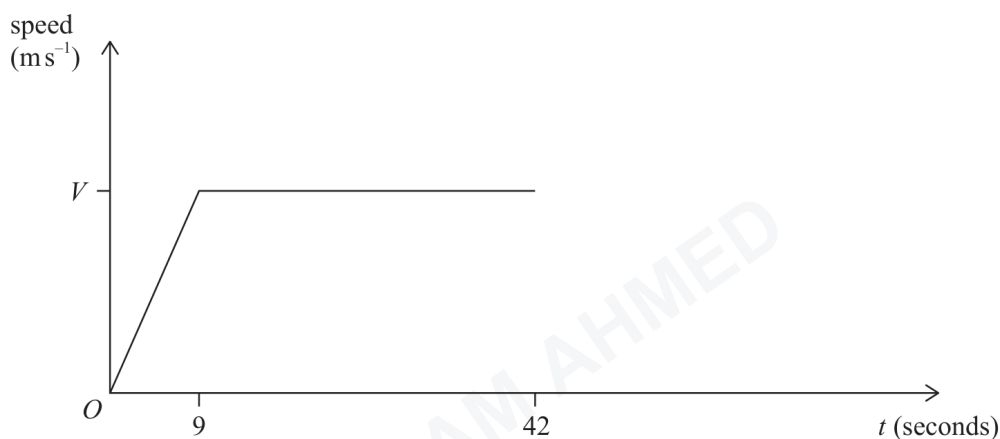


Figure 2

Figure 2, on page 11, shows the speed-time graph of a cyclist P , for his journey from A to B .

At time $t = 0$, P starts from rest at A and accelerates uniformly for 9 seconds until his speed is $V \text{ m s}^{-1}$

He then travels at constant speed $V \text{ m s}^{-1}$

When $t = 42$, cyclist P passes B .

Cyclist P continues to cycle along the road in the same direction at the same constant speed, $V \text{ m s}^{-1}$

When $t = 6$, a second cyclist Q sets off from A and travels in the same direction as P along the same road. She accelerates for T seconds until her speed is 3.6 m s^{-1}

She then travels at constant speed 3.6 m s^{-1}

Cyclist Q catches up with P when $t = 54$

(d) Find the value of T

(5)

5. A car travels at a constant speed of 40 m s^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 m s^{-2} for 15 s and then accelerates at 1 m s^{-2} for a further T seconds until he catches up with the car.

- (a) Sketch, on the same axes, the speed-time graph for the motion of the car and the speed-time graph for the motion of the motorcyclist, from time $t = 0$ to the instant when the motorcyclist catches up with the car.

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5. A car travels at a constant speed of 40 m s^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 m s^{-2} for 15 s and then accelerates at 1 m s^{-2} for a further T seconds until he catches up with the car.

At the instant when $t = t_1$ seconds, the car and the motorcyclist are moving at the same speed.

- (b) Find the value of t_1

(2)

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5. A car travels at a constant speed of 40 m s^{-1} in a straight line along a horizontal racetrack. At time $t = 0$, the car passes a motorcyclist who is at rest. The motorcyclist immediately sets off to catch up with the car.

The motorcyclist accelerates at 4 m s^{-2} for 15 s and then accelerates at 1 m s^{-2} for a further T seconds until he catches up with the car.

- (c) Show that $T^2 + kT - 300 = 0$, where k is a constant to be found.

(6)

DR ESAM AHMED

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.
- They both start from rest, leaving the start line at the same time.
- Ajaz accelerates at 0.8 m s^{-2} up to a speed of 4 m s^{-1} and then maintains this speed until he crosses the finish line.
- Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.
- Ajaz and Beth cross the finish line at the same time.
- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line.

(3)

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Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (b) Find the time taken by Ajaz to complete the race.

(4)

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- (c) Find the value of T

(4)

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Ajaz and Beth cross the finish line at the same time.

- (d) Find the difference in the speeds of the two children as they cross the finish line.

(2)

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